

Tactile sensing lets quadruped robots carry shifting loads

Tactile sensing that enables quadruped robots to sense shifting weight could expand their role from labs to real-world applications, including warehouse handling, disaster



Quadruped robot carrying a shifting load (Credit: Carnegie Mellon University Mechanical Engineering)

response, and field logistics. By detecting load movement researchers at Carnegie Mellon University (Pittsburgh) have demonstrated that by real-time adjustment of gait instantly, the system enables robots to carry loose or unstable objects without dropping them. The advance improves balance, safety, and au-

tonomy, making legged robots more practical for unpredictable, human-centric environments.

Flexible sensors generate energy for next-gen electronics from pressure

A new class of self-powered, flexible tactile sensors that generate energy from pressure could accelerate smart wearables, interactive devices, and secure human-machine interfaces by eliminating bulky batteries. Researchers at Anhui University (Hefei, China) have developed MXene-based 'one-body' units that sense touch, store energy, and map pressure in compact arrays with fast response times, enabling high-accuracy gesture recognition and pressure-controlled functions in health monitoring, adaptive interfaces, and next-gen wearables.

2D wafer bonding joins atom-thin layers for next-gen devices

A direct 2D wafer bonding method that joins atom-thin semiconductor layers without glue-like materials could accelerate smaller, faster electronics for advanced transistors, sensors, and high-performance chips. Researchers at Songshan Lake Material Laboratory, Chinese Academy of Sciences, have enabled precise control of layer count and twist angle for materials. Cleaning interfaces and controlling layer stacking would allow engineers to build precise heterostructures (for instance, $\text{MoS}_2/\text{MoSe}_2$) with fewer defects, boosting performance, and reducing device footprints for next-gen computing, IoT, and wearable systems.

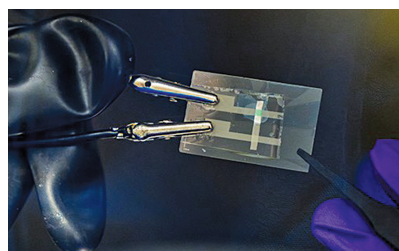
Maritime IoT network tested with up to 35km range

A first-of-its-kind real-world maritime IoT network has proven low-power, long-range ocean connectivity, unlock-

ing continuous marine data feeds for safety, weather, navigation and environmental monitoring at sea. Korean researchers at the Electronics and Telecommunications Research Institute (ETRI) completed the first real-world test of a Maritime Internet of Things (MIoT) network. Sensors on ships, buoys, and ports maintain stable links up to 35km. They can scale to hundreds of devices, enabling denser networks for aquaculture, small-craft tracking, ecological monitoring, and coastal infrastructure management.

Fully stretchable OLED displays for curved, soft, and moving surfaces

Researchers at the University of Chicago's Pritzker School of Molecular Engineering developed fully stretchable OLED displays that conform



Next generation stretchable OLED screens (Credit: UChicago Pritzker School of Molecular Engineering/Jason Smith)

to curved, soft, and moving surfaces, bringing them closer to reality, using a new cathode and an elastic polymer stack that remains bright and conductive even when stretched. This breakthrough points

to skin-like screens for wearable health patches, interactive soft robotics, adaptive consumer interfaces, and medical diagnostics, enabling displays that won't crack under strain while maintaining performance similar to rigid OLEDs.

Satellite technology connects remote areas for conservation

At Malaysia's Al-Sultan Abdullah Royal Tiger Reserve in Ulu Tembeling, Pahang, satellite-based connectivity is



EdgePoint and Pahang State Parks Corporation Launch Connectivity for Conservation at the Al-Sultan Abdullah Royal Tiger Reserve (Credit: <https://edgepointinfra.com>)

bringing internet and power to remote conservation zones, like the Al-Sultan Abdullah Royal Tiger Reserve, enabling real-time communication for rangers and researchers tracking wildlife, reporting incidents, and coordinating responses. By linking guard posts and stations via Low

Earth Orbit links with solar/battery power, the network supports faster anti-poaching action, emergency coordination, and data-driven habitat protection, overcoming terrain and infrastructure limits.